

Rifampicin binds tightly to an RNA polymerase molecule close to its active site. This affects the activity of the enzyme.

- (b) (i)** RNA polymerase catalyses the formation of messenger RNA (mRNA) from DNA.

State the term for this process.

..... [1]

- (ii)** During the formation of RNA, a number of events occur that involve the action of RNA polymerase.

Suggest ways in which rifampicin can affect the activity of RNA polymerase.

.....
.....
.....
.....
.....
.....
.....
.....
..... [3]

- (c) RNA polymerase is composed of five different polypeptides. Gene *rpoB* codes for one of these polypeptides known as the β -subunit.

One or more mutations in a specific region of *rpoB* result in strains of *M. tuberculosis* that are resistant to rifampicin. In these strains, mutations often occur in two DNA triplets in this region, in positions 526 and 531.

Table 2.2 summarises the results of an investigation into seven rifampicin-resistant strains, **A** to **G**, that have amino acid changes for positions 526 and 531.

Table 2.2 includes:

- the change in the **mRNA codon** for position 526 or position 531
- the amino acid change that has occurred as a result of the mutation
- the minimum concentration of rifampicin required to inhibit growth of the bacterial strain (MIC)
- the number of **other** mutations occurring within the specific region of *rpoB*.

Table 2.2

Key

$\approx$ approximately

$\geq$ greater than or equal to

$\leq$ less than or equal to

strain	codon involved	mRNA codon change	amino acid change	MIC / $\mu\text{g cm}^{-3}$	number of other mutations in the specific region
A	526	CAC $\rightarrow$ UAC	His $\rightarrow$ Tyr	≤ 50	0
B	526	CAC $\rightarrow$ AAC	His $\rightarrow$ Asn	≥ 100	1
C	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	$\approx 50\text{--}75$	2
D	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≥ 100	3
E	526	CAC $\rightarrow$ CGC	His $\rightarrow$ Arg	≈ 50	3
F	526	CAC $\rightarrow$ UUC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUG	Ser $\rightarrow$ Leu		
G	526	CAC $\rightarrow$ UAC	His $\rightarrow$	≥ 100	3
	531	UCG $\rightarrow$ UUC	Ser $\rightarrow$ Phe		

(i) Complete Table 2.2 to show the amino acid changes that have occurred in strains **F** and **G**.

[1]

(ii) With reference to Table 2.2, list the strains of *M. tuberculosis* that show the greatest resistance to rifampicin.

..... [1]

(iii) Suggest reasons to explain why strains **C**, **D** and **E** show:

- resistance to rifampicin
- different levels of resistances to rifampicin.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [3]

[Total: 13]